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ASTS as method: towards a critical practice

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Abstract

This contribution advocates for reconceptualizing Art, Science and Technology Studies (ASTS) as method rather than merely an interdisciplinary convergence. Rooted in the historical notion of *technê* – centring knowledge formations grounded in practice – the paper presents method as an active, performative mode of engaging with and shaping technoscientific realities. In dialogue with Science and Technology Studies (STS), especially its refusal to separate theory, method and practice, ASTS is positioned as extending this tradition by incorporating the epistemic, aesthetic and material capacities of artistic practice. Rather than observing from a distance, ASTS adopts a critical stance and intervenes within socio-technical systems, actively shaping them while highlighting the situated, contingent and political aspects of knowledge production. By presenting methods and case studies that embody this orientation, the contribution emphasizes how ASTS aims to work generatively with complexity – reconfiguring the infrastructures, imaginaries and modes of critique that shape contemporary technoscientific life.

Keywords

critical ASTS; technical reconfiguration; hybrid practice; method assemblage; situated knowledge production

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Los ACTS como método: hacia una práctica crítica

Resumen

*Esta contribución aboga por reconceptualizar los estudios de arte, ciencia y tecnología (ACTS) como método en lugar de considerarlos simplemente una convergencia interdisciplinar. Arraigado en la noción histórica de formaciones de conocimiento centradas en la técnica (*tékne*) basadas en la práctica, el documento presenta el método como un modo activo de interactuar y dar forma a las realidades tecnológicas. En el diálogo con los Estudios de Ciencia y Tecnología (CTS), especialmente su negativa a separar la teoría, el método y la práctica, los ACTS se posicionan como una extensión de esta tradición al incorporar las capacidades epistémicas, estéticas y materiales de la práctica artística. En lugar de observar a distancia, los ACTS adoptan una postura crítica e intervienen dentro de los sistemas sociotécnicos, definiéndolos activamente a la vez que destacan los aspectos políticos, contingentes y situados de la producción de conocimiento. Al presentar métodos y casos prácticos que encarnan esta orientación, la contribución enfatiza cómo los ACTS tienen como objetivo trabajar generativamente con la complejidad, reconfigurando las infraestructuras, imaginarios y modos de crítica que dan forma a la vida tecnológica contemporánea.*

Palabras clave

ACTS crítico; reconfiguración técnica; práctica híbrida; ensamblaje de métodos; producción de conocimientos localizados

Introduction: from convergence to intervention

The convergence of art with science and technology studies, formalized as ASTS, has given rise to a growing field of inquiry that serves as a fertile ground for new forms of practice. This convergence can be formulated as more than an interdisciplinary amalgam. By contributing to recent scholarship that positions ASTS as an emerging epistemic formation with distinct methodological orientations (Rogers *et al.* 2021, Salter *et al.* 2017, Calvert & Schyfter 2017), this paper aims to make a case for ASTS as method. It does so by addressing the increasingly complex entanglement of technoscientific advancements and daily life through critical artistic perspectives. At the core of this orientation lies a strong commitment to practice as a space for intervention and knowledge production. The concept of *technê* initially highlights a historical understanding of knowledge as inseparable from the material conditions and practices through which it is enacted. Such an understanding is particularly relevant today, where technoscientific object-processes – especially intelligent machines – are created and maintained by corporate actors that refuse to recognise how they are shaped by situated and contingent processes. Beyond a historical starting point, *technê* functions as a structuring logic that enables us to rethink what methods, in their openness, can achieve – not as rigid sets of strategies and techniques but as integral ways to attend to how technoscientific realities are formed and contested, and to explore further possibilities. This issue becomes especially salient within ASTS, where methodological propositions – such as critical-technical practice and critical making discussed here – demonstrate that the goal is not merely to analyse from a distance.

The case studies that follow articulate situated instances of taking a critical stance towards the innovative trend of human-machine convergence and to tease out its politics. They demonstrate how such hybrid practices – through speculative and disruptive approaches – influence the production and circulation of technoscientific knowledge, foregrounding methods as performative constructs that enact the very realities they aim to examine. Incorporating artistic practice into this performative framework extends this proposition by going beyond illustrating or critiquing technoscientific systems through participating in their ongoing reconfiguration. The treatment of theory, method and empirical engagement as inseparable – a stance taken up by science and technology studies (STS) – is thus hybridized with the epistemic, material and speculative affordances of artistic practice. This reformulation then becomes a generative methodological formation – one capable of reshaping the infrastructures, imaginaries and modes of critique through which technoscientific worlds are made and remade.

1. *Technê* and its legacy: situated knowledge production

The intersection of technoscience that conjoins science and technology to intertwine scientific and technological facts and artefacts with artistic practice, had long been lurking under the radar before such a formal designation emerged.[1] ASTS adds another layer of hybridization to STS as an already interdisciplinary field that examines how technical

1. For a concise historical overview, see Chris Salter's "The Art-Science Complex" in Rogers *et al.*

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infrastructures affect social structures and how social concerns feed into the formation of technical object-processes. It aims to intersect the materiality, situatedness and embodied dimension of artistic practice with technoscience, asserting that "artists are practising STS by material means" (Rogers 2021, 231), while asking "how might methodologies from art and design practice be introduced into STS?" (Forlano & Sedini 2021, 198). As innovative as this conglomeration may seem, the strict separation between art, science and technology that later became a matter of fact, was not a natural or inherent division. The enigmatic term *technê* emphasises that after all, interdisciplinary thinking became prevalent in the aftermath of systematic institutionalization of knowledge in the form of disciplines. The cultural division between science, technology and social concerns is therefore not natural but historically contingent.

In ancient Greek, *technê* was a contested term, difficult to define and delineate within our current neatly divided knowledge systems. While tying it to craft and practice as opposed to dialogical reasoning is the common understanding of the term, these distinctions were not so clear-cut and self-evident. Knowledge boundaries were porous, permeable and constantly shifting. *Technê*, however, was about forms of knowledge that resonated with practice, spanning craft, such as carpentry and painting, to sciences, such as medicine and arithmetic. *Logos* or dialogical reasoning could also be practiced as part of *technê*, bringing it closer to science. To complicate matters further, Plato, who shaped the early understanding of the term, used *technê* and *epistêmê* inseparably as lines of expertise. Meanwhile, the consensus was that *technê* "was fundamentally about how to do things, 'knowing-how rather than simply knowing-that'", a notion associated with scientific knowledge or *epistêmê* (Schatzberg 2018, 18). In his historiographic volume *Technology: Critical History of a Concept*, Eric Schatzberg titles the second chapter "The Trouble with Techne" to highlight the broad scope of *technê*, which resists our current tendency to taxonomize forms of knowledge production. *Technê* is practice itself, but also the kind of knowledge that emerges from this practice, which at times requires reasoning to incorporate *logos* into the process.

But as ecologist Monica Gagliano (2024) insists, *technê* has always been about the "art and science" of practice, not merely straightforward pragmatic means. Zipping through time, multitudes of paradigm shifts that have occurred between these two moments, from Plato to contemporary times, makes evident the oscillation between *technê*, *epistêmê*, and later the Latin concept of *ars* to meld knowledge spaces. But technology as we know it today was nowhere in this cycle. If the radical separation between *technê* and *epistêmê* originated from Aristotelian thought, the very concepts and practice of technology bridge this chasm to exhibit the applied traction of science (Schatzberg 2018). To avoid a lengthy historical detour and keep a modern focus, if technology represents "the application of science to practical ends" (Salomon 1984, 119), and at the same time the products resulting from such practices, then it is the sharp edge of industrial acceleration that gave technology such a narrow focus to defy the harmonizing spectrum of *technê* and *logos*.

In response, the concept of technoscience conjoins science and technology, demonstrating how scientific and technological facts and artefacts do not emerge in isolation but are co-produced through complex networks of people, institutions and material infrastructures – all shaped by power and politics. We are thus already getting one step closer to the spirit and essence of *technê* that does not adhere to strict taxonomies and distinctions. Bruno Latour (1999) writes: "there is nothing that we can define philosophically or sociologically as an object, as an artifact or a piece of technology" (190-191). He is, of course, critical of the tendency towards compartmentalization and of boxed-in thinking associated with technology as an entity devoid of relations and attachments. Derived from the broader framework of actor-network theory (ANT),[2] technology is always already a reduction of a whole array of actors, things, and relations and their interconnections that are socially, culturally and politically charged. These relations are then further shaped by what ANT calls "translations", which describe the processes through which loops of transformations shape the production and dissemination of knowledge. Translations are intrinsically a series of displacements, producing and disseminating knowledge while overlooking the complexities involved, such as discrepancies, power dynamics, questions of labour and contingencies (Callon 1984); issues that *technê*, grounded in human-centred practices, has long grappled with. As a result, technoscience points to *technê* and its legacy to examine what it means to embed knowledge production within social and relational contexts to think of messes, permeable borders and uncertainties.

But beyond technological operations, STS also scrutinises the bold claims and normative narratives constructed by the main producers of such technical objects, systems and infrastructures – in other words, the big tech. It does so by examining the lineage of technical objects and processes, exposing their blind spots and tracing their genealogies. In demanding that these claims be revisited, STS insists on situating the technical within the messiness of the world – including social realities. Returning to the intersection of arts, science and technology, *technê* has already shown that engaging with both the craft of technoscience and the craft of the arts – along with the complexities such an inquiry entails through practice – is neither strange nor unnatural. However, carrying over the spirit of *technê* across time and into contemporary academic cultures carries with it discord and dissonance. It requires navigating shifting contexts, onto-epistemologies and methods that have shaped these fields. In the spirit of *technê*, the centrality of practice becomes the site that can hold these inquiries together to bring technology back – or rather, down – to earth, considering how it can be situated and embodied within a broader framework of people, things,

2. Developed as both a methodological and theoretical framework, ANT treats humans and non-humans horizontally, attributing agency to things. For further context, see Latour's *Reassembling the Social*.

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actions and relations. Practice can be formulated as the crux of applied forms of knowledge production across disciplines to think about why practice in the art studio can have affinities with practice in the science lab (Rheinberger 2019) – something that, in fact, *technê* prominently underscored. ASTS thus becomes a means to make these complex interrelations that STS has been concerned with and discursively pointed out publicly tangible. As art historian Caroline Jones writes in the foreword for the ASTS Handbook, "what the emergence of . . . ASTS makes clear is that science has failed to produce anything like the public sphere of art to entertain the cultural and ethical arguments we need to have about our technoscience" (Jones 2021, xix).

2. Zooming in: ASTS as critique

One domain deserving broader public attention is the entanglement of life with intelligent machines now as infrastructural agents that shape social norms, political processes and modes of attention and interaction, demanding new ways of understanding agency and relationality. ASTS's approach to this domain is afforded by its mutable stance and formal autonomy – distinct from the homogeneity required for other types of hierarchical knowledge production. In this context, ASTS can embrace a posthumanist view of the world and a grounded understanding of human-machine convergence, framed as collaborations between bio and techno materialities in specific, situated scenarios.

Artistic practice is here the component that enables these material interactions to take shape and assume "speculative realities", where possible futures allow for the space of imagination to unfold. This affordance stems from the materiality, representations that sit outside the quotidian experience and social engagement within a public setting (Borgdorff *et al.* 2020, Sormani *et al.* 2020). Artistic practice thus holds the potential to generate aesthetic, poetic and performative experiences that traverse complex concepts, addressing the problem of imagination and loosens not only the rigid bounds across disciplines but also the normative ways of understanding the world. As Chris Salter (2023) notes, this potential encompasses "plants, animals, cells, bacteria, viruses, physico-chemical phenomena and machines framed by larger ecological-technological contexts" (337). Machine agency has, in fact, increasingly become a site of contemplation, if not tension – across schools of thought due to its unique capacity for action that is notoriously perceived as autonomous and hence placed alongside that of humans. ASTS practices that galvanize this juncture, open a literal space for their existence to build imaginaries around. As such, "they wholeheartedly blur and hybridize distinctions between organic and nonorganic, living and nonliving, and generate complex amalgams of technical-social-cultural-material-semiotic actions and objects" that in effect take place as they perform in front of our eyes (Salter 2015, 6).

The spectrum of ASTS practices thus demonstrates the possibilities of making and doing human-machine sensibilities to grasp the theorisation of everyday practices within technoscience. On another level, while artistic practice has long presented these transformative experiences by wild juxtapositions, appropriations, and the adoption of a broad array of imaginaries and materialities on an aesthetic level, it has also played an integral role in cultural, social and political critique. In this vein, artistic practice foregrounds problem-oriented perspectives that accentuate the discomfort caused by stark contrasts across multiple levels of reality that contemporary subjects must navigate in an increasingly complex world (Kester 1998, Rancière 2006). The socio-technical domain is one such multiplex reality that is becoming an integral part of daily life. Concurrently, within STS, human-machine hybridities as socio-technical systems are the subject of extensive analysis (Hayles 1999, Latour 1993, Suchman 2007), bringing forward perspectives that challenge uncritical celebration of technological innovation. While ASTS can position technoscientific practices within their broader socio-material contexts through practice, its capacity to go beyond showcasing the affordances of technical systems and towards active critical engagement sets it apart from the mere juxtaposition of art, science and technology towards novel compositions.

Rather than affirming technoscientific innovation through aesthetic demonstrations, ASTS can foreground methodologies that critically interrogate the integration and deployment of technologies within lived social contexts. Such a critical entanglement calls for what John Law (2004) refers to as generous method – approaches that go beyond normative methods to address contingencies and irregularities arising from hybrid inquiries.

Two examples that can activate such a space are critical-technical practice (CTP), which comes from AI research, and critical making, rooted in design. Both methods instantiate *technê*'s legacy as knowledge grounded in making, where materiality, reflexivity, and embodied insight converge to create forms of understanding irreducible to abstract logic or detached observation.

Widely adopted within art and design as a result of interdisciplinary research, CTP proposes a situational and interactional model by suggesting an uncomfortable ground of practice "one foot planted in the craft work of design and the other foot planted in the reflexive work of critique" (Agre 1997). CTP has been applied within interaction design to encourage reflective engagement with technology, addressing implicit values and assumptions embedded in design processes and artefacts (Sengers 2005). It has also expanded into design-oriented community work, highlighting how artists and designers adopt CTP-inspired strategies to foreground alternative value frameworks and facilitate critical engagement with technology within marginalized or excluded communities (Vertesi *et al.*, 2017). These adaptations – along with other appropriations of the concept within media and technological domains that challenge the status quo of technical practices and propose alternative formulations (Soon & Velasco 2023) – demonstrate how CTP has evolved beyond its disciplinary origins in AI to influence a wider range of artistic and design-led interdisciplinary projects. Another established pathway that comes from design research

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is critical making as a method of collective practice involving theoretical reflection and experimental making, followed by critical assessment and experimentation with the work to "achieve value through the act of shared construction, joint conversation, and reflection" (Ratto 2011). Critical making is increasingly employed as an approach that places material engagement at the centre of critical inquiry. Extending Ratto's original framework into artistic contexts, Garnet Hertz (2012) underscores critical making's role as both a pedagogical instrument and a practice-based research method in design education and media art. The method can therefore be adapted as a bridge between theoretical critique and tangible practice, being able to confront and interrogate the social and political dimensions of technological artefacts.

3. Zooming out: STS and ASTS as method

But to adopt a more integrative framework beyond focusing solely on methods within research, by circling back to STS and its socio-material foundation, it becomes evident that unlike traditional disciplinary frameworks that STS draws from, as Law (2017) argues, the field "rolls theory and method and empirical practice together . . . and insists that they are all part of the same weave and cannot be teased apart". In this respect, STS considers theories and methods in situated practices and studies the effects of these practices; in this sense, it is not only a study of practices but also "it is practice itself". Building on this integrated notion, Law's proposition of "STS as method" further clarifies this inseparability by highlighting methods as performative rather than merely descriptive; for Law, methods actively enact and produce the realities they study. By positioning method as inherently situated, embedded in relational contexts, and generative of specific socio-material arrangements, Law shifts the focus from observing socio-technical phenomena to actively intervening and shaping them. The rolling together becomes evident in "the weave that links methods together with subjects, objects, expressions and representations, and institutions" to think at once of methods embedded within object-processes and methods of studying them through STS.

To shift this perspective and focus it on ASTS, methods such as CTP and critical making help articulate how critique and creation are inseparable. Both approaches incorporate reflection into practice, demonstrating how ASTS performs as method. By adopting these methods, ASTS not only enhances reflexive awareness of its interventions but also actively constructs new pathways for engaging socio-technical complexities. Such methodological orientations render ASTS a dynamic, interventionist mode of inquiry rooted in practice and committed to critical engagement with the contingent, embodied, and situated processes that shape technoscientific worlds. This understanding of method as a performative, inseparable element of socio-material practices provides a productive framework to conceptualize ASTS not just as a novel interdisciplinary hybrid but as method itself. This provocation resonates with and surpasses the aim of its proponents to "propose ASTS as a theoretical and methodological framework" (Rogers & Halpern 2021, 41). ASTS as method thus highlights the generative capacity of critical ASTS practices to interrogate and actively reconfigure the relational and material conditions within which technoscientific phenomena are produced, maintained and transformed.

4. ASTS in action: case studies

In the spirit of STS where case studies guide inquiry, three cases framed as instances of critical ASTS demonstrate how it can operate as method, critically questioning and actively intervening in the networks of relations and material practices that give rise to and sustain technoscientific phenomena.[3] Spawning AI's *Have I Been Trained?*[4] platform exemplifies what it means to enact ASTS as method. Instead of just analysing and criticising the ethics of large datasets used to train generative machine learning models, such as image generators, from a discursive distance, the platform performs an infrastructural intervention directly within the technoscientific pipeline. Spawning AI enables artists and creators to search for their work in major AI training datasets and request removal, offering a rare layer of consent and control in a domain otherwise dominated by opaque systems and large-scale data harvesting (Wiener 2023). By designing a consent-based interface between artists and machine-learning systems, *Have I Been Trained?* goes beyond mere commentary – it practically redefines the terms of participation in human-machine collaborations. This intervention aligns with the core of ASTS as method, which understands method as a performative, world-making practice. The project is therefore not just a technical tool as it reshapes the socio-material relations between artists, datasets, platforms, and audiences. It makes visible and actionable the often-invisible infrastructures governing AI development and distributes agency throughout that system. This kind of methodological intervention, integrated into technological processes yet a publicly engaged socio-political act, demonstrates the capacity of ASTS to reconfigure how technoscientific realities are constructed, engaged with, and contested.

In a similar vein, Claudia Larcher's *AI and the Art of Historical Reinterpretation*[5] mobilizes AI-generated imagery to contest the gendered

3. While ASTS often engages the art-science tradition – most notably in bioart – the case studies here focus on art and technology, reflecting this contribution's emphasis on human-machine entanglements.
4. https://spawning.ai/have-i-been-trained
5. www.claudialarcher.com/work/theartofhistoricalreinterpretation

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omissions and biases embedded in historical image archives. Rather than reinforcing existing power structures through uncritical AI applications, the project suggests an alternative: using AI's generative capacity to fill historiographic gaps by creating a speculative and inclusive visual record. In doing so, Larcher critiques the underlying politics of dataset composition, highlighting how technical systems absorb and perpetuate exclusionary narratives. The growing fictional image archive, designed to circulate across digital platforms, acts as both critique and intervention – reconfiguring the material that future models could passively inherit. Within the context of ASTS as method, the project exemplifies how artistic inquiry can rework the socio-technical infrastructures of knowledge production, challenging the presumed neutrality of AI while staging alternative futures through data-driven media. Larcher's project demonstrates how the speculative edge of artistic practice can do more than aesthetic-poetic explorations; it can bring multiple realities into perspective and imagine a future otherly built.

While the first two examples focus directly on generative AI and its training datasets, the third one zooms out to question systems of datafication and algorithmic control. Tega Brain and Surya Mattu's *Unfit Bits*[6] exposes the sociotechnical logics behind biometric self-tracking and its intertwinement with surveillance capitalism. Beyond a discursive critique, the project suggests absurd "hacks" for generating activity data – such as strapping a fitness tracker to a metronome or bicycle wheel – satirically responding to life insurance companies that offer discounted premiums based on access to applicants' smart wearable data.[7] In doing so, *Unfit Bits* conducts a methodological intervention that challenges the assumed objectivity of biometric data, exposing the fragile link between the messiness of bodies, algorithmic interpretation of biometric data, and regimes of discipline and control. The project exposes the coercive infrastructures behind wellness culture and quantification, linking them to broader systems of risk assessment, behavioural governance, and algorithmic bias. *Unfit Bits* embodies a subversive refusal that shows how ASTS can render power visible and contestable through performative disobedience enacted on the very technical systems it critiques. As a result, it unravels an epistemological intervention that opposes the status quo of the big tech and shifts how their products are perceived, understood and experienced.

Conclusion: method assemblage

What begins as a historical and conceptual tracing of *technê*, unfolds into an argument for viewing ASTS as method. This shift is rooted in STS's legacy of refusing to separate theory, method, and empirical engagement, and more pointedly, in its view that methods are not neutral instruments of observation but actively influence what they study and how to study. ASTS inherits and extends this proposition, but does so by integrating the epistemic, aesthetic and material affordances of artistic practice – its capacity to make public, to speculate, to disrupt and to embed critique in action. In particular, the incorporation of methods like CTP and critical making signals a commitment to interrogating technological systems through direct, situated, and critical intervention. These frameworks demonstrate how ASTS enacts method by entangling critique and creation, concept and form, systems and publics.

The projects discussed make this methodological orientation tangible by operating within technoscientific systems. They exemplify ASTS's methodological proposition: that it is both possible and necessary to craft new modes of engagement where critique is not separated from making, and where methods serve as sites of intervention to redefine the boundaries of the field, and ultimately how this attitude can be regarded as method in its own right.

In conceptualizing ASTS as method, the goal was to emphasise the need for approaches that are materially grounded, politically attentive and epistemologically open-ended. The idea goes beyond developing new tools to analyse technoscientific phenomena to instead reshape the terrains on which such phenomena occur through Law's generative method. Moving beyond a strict formulation, generative methods think of ways to expand the periphery of normative methods. One way of framing this expansion is through method assemblage (Law 2004) – a way of bringing together multiplicity and heterogeneity of sites of inquiry together, in this case, the convergence of art, science and technology. Method assemblage is the means by which the conceptualisation of ASTS as method becomes possible, as Law proposes, by way of "crafting of relations that shape, mediate and separate an object in-here, its relevant context out-there, and then an endless set of out-there relations, processes and all the rest that are a necessary part of the assemblage but at the same time have disappeared from it" (84). By foregrounding artistic practice as a constitutive element of knowledge production, ASTS mobilizes method assemblage to entangle theoretical critique, material experimentation, and most importantly, socio-political intervention, since "art that engages with science and technology is ultimately political" (Rogers 2021, 407). It does so by crafting and re-crafting the very relations – between humans and machines, between histories and their erasures, between infrastructures and publics – that give technoscientific systems their shape and meaning. The convergence of art, science, and technology is thus not only a mode of collaboration across disciplines but a methodological configuration that activates the multiplicity of technoscientific realities. ASTS as method does not aim to reduce complexity but to work with it – to hold together disparate modes of knowing and making, and to open up speculative, critical and transformative engagements with the socio-technical world. Integrating all these seemingly disparate ingredients of knowledge production and viewing it as method ultimately reflects the spirit of *technê* and its heterogeneous incoherence as

6. http://www.unfitbits.com/
7. The outline of the insurance premium reward scheme can be accessed at https://money.com/life-insurance-fitness-rewards/ as referenced by Brain: https://tegabrain.com/Unfit-Bits

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the essence of practice. *Technê* not just underpins ASTS as a method but also animates its refusal to disengage making from thinking and knowledge from context. It invites us to think with, through and against dominant technoscientific imaginaries, offering not only a critique of how knowledge is produced but also new ways of making it otherwise.

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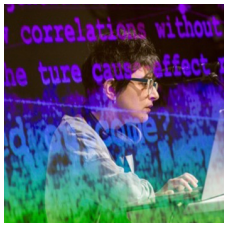
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Refereed single-authored Publications

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Artistic Contribution

Ars Electronica, Linz, Austria (2023-2024-2025); Barents Spektakel Festival, Kirkenes, Norway (2025); Bergen Center for Electronic Arts, Bergen, Norway (2025); Radio Worm, Worm Rotterdam, Netherlands (2025); IMPAKT Center for Media Culture, Utrecht, Netherlands (2024); New York City Electroacoustic Music Festival (2024); Fylkingen New Music & Intermedia Center, Stockholm, Sweden (2024); Hessian Center for Artificial Intelligence, Darmstadt, Germany (2023); Kunsthal Extra City, Antwerp, Belgium (2022); Body Electric Retrospective 2014-2020, Toronto, Canada.

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